

Kitty Number

Tags:

Time Limit: 1 s Memory Limit: 64 MB
 Submission: 13 AC: 5 Score: 99.71

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Codes

AI Code Tips

Description

A Kitty Number N is a positive integer satisfies that given two any positive integers A and B, and among A, B and N, if we have

$$N \mid ((A^2) * B + 1)$$

Then

$$N \mid (A^2 + B)$$

Now, you are asked to judge a given positive number is Kitty Number or not.

Input

The first line there is a number T (0 < T < 5000), denoting the test case number.

The following T lines for each line there is a positive number N (0 < N < 5000) you need to judge.

Output

For each case, output “YES” if the given number is Kitty Number, “NO” if it is not.

Samples

input	Copy
2 3 7	
output	Copy
YES NO	

Hint

X | Y means X is a factor of Y, for example 3 | 9;
 X^2 means X multiplies itself, for example 3^2 = 9;

$X*Y$ means X multiplies Y, for example $3*3 = 9$.

Source

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